Autonomous Drone Security Monitor Project Synopsis

Team Member Names

- Roberto Manzano
- Tyler Lorentz
- Andrew Narlock
- Bryan Nguyen
- James Fischer

Breakdown of Work Completed by Each Team Member

Roberto Manzano

- Assisted with anomaly detection implementation and testing
- Helped debug telemetry processing and application functionality
- Contributed to unit testing and quality assurance
- Assisted with project documentation and deliverables
- Participated in GitHub version control and code integration

Tyler Lorentz

- Led project architecture and overall system design
- Coordinated development efforts and feature integration
- Implemented major application components
- Assisted with database and backend functionality
- Participated in GitHub version control and code integration
- Integrated SQLite database functionality into the application

Andrew

- Developed and improved graphical user interface components
- Assisted with dashboard implementation
- Contributed to user interaction features and testing
- Participated in GitHub version control and code integration

- Contributed to unit testing and quality assurance

Bryan

- Assisted with telemetry simulation functionality
- Helped implement monitoring features
- Contributed to testing and debugging activities
- Participated in GitHub version control and code integration
- Designed and implemented the Database Manager component
- Integrated SQLite database functionality into the application

James

- Assisted with documentation and project organization
- Helped manage user stories and project planning
- Participated in testing and validation efforts
- Participated in GitHub version control and code integration

Total Hours Contributed

The following hours were tracked using YouTrack throughout the project:

Team Member	Hours Contributed
Roberto Manzano	45
Tyler Lorentz	62
Andrew	31
Bryan	29
James	49

Problems We Had to Overcome

One of the primary challenges was coordinating development among multiple team members using GitHub. Because several developers worked on related components

simultaneously, merge conflicts occasionally occurred and required careful review before integration.

Another challenge involved integrating telemetry generation with anomaly detection. The system needed to correctly identify issues such as low battery conditions, unsafe maneuvers, and abnormal GPS behavior while minimizing false alerts. Multiple rounds of testing and refinement were required to achieve reliable results.

The team also spent significant time ensuring communication between application components, including telemetry updates, anomaly logging, database persistence, and dashboard visualization. Debugging these interactions required collaboration and extensive testing.

Project Shortcomings

While the project successfully satisfies the primary requirements, there are several areas that could be expanded in future versions.

The anomaly detection system currently relies on predefined thresholds rather than advanced predictive analytics. The dashboard visualization could be improved with a more sophisticated map-based display and additional data visualizations. Database reporting and query capabilities could also be expanded to provide more detailed historical analysis.

These limitations do not prevent the application from meeting project requirements but represent opportunities for future improvement.

Items That Should Be Considered for Extra Credit

The team believes the following features should be considered for extra credit:

- Enhanced anomaly detection beyond basic low battery and GPS monitoring
- Persistent anomaly logging and historical record keeping
- Real-time dashboard updates and monitoring interface
- Additional telemetry processing capabilities
- Expanded testing and validation efforts
- Improved user experience through graphical visualization and alert management

These additions extend the project beyond the minimum functional requirements and improve overall usability, maintainability, and system quality.